

Lesson 5: Is there a similar mathematical relationship between the masses and the changes in velocity for two objects in any other collision conditions?

Navigate

With your class



1. What were we wondering about this mathematical relationship at the end of our last class? Reference slide.

Plan an investigation

With your class



2. Orient to a simulation that we can use to help us collect data on some of these other collision conditions.

Carry out an investigation

With a partner



3. Use the simulation to test at least one elastic collision and one inelastic collision: <http://openscienced-static.s3.amazonaws.com/HTML+Files/Inelastic+and+Elastic+Collisions.html>
4. For each collision record the following:
 - The type of collision
 - The mass of each cart
 - The starting velocities of each cart
 - The final velocities of each cart

Test our model

With a partner



5. Does the mathematical relationship we discovered for Collisions E and F hold for none, one, or both of the collisions you tested in the simulation?
6. Use a white board to document your mathematical thinking for this.

Use mathematical thinking

With a partner



7. Try to simplify the mathematical relationship we figured out so that it no longer has fractions. What can you do to move the variables out of the denominators?

Recalling other conserved quantities

Turn and talk



8. Based on our work in prior units, what are some other physical quantities that are conserved?

Use mathematical thinking

With your class



9. Prepare to develop a geometric way to visualize the quantity that is conserved ($m \cdot \Delta v$)

Turn and talk



10. What are some shapes you've used in the past to represent the quantity that is the product of two variables ($a \cdot b$)?

With your class



11. A 3 kg cart is stationary before a collision and it had a speed of 4 mph after the collision.

- How could we use a rectangle to represent the amount of momentum change this object experiences?

Evaluate the geometric model

With your class



12. What did the geometric representation help us visualize?

13. What were some of the limitations of the geometric representation?

Use mathematical thinking

- $m \cdot \Delta v$ is change in momentum.
- It has two variables as well as a Δ symbol.
- Let's consider what momentum is, by recalling how we determine the change (Δ) for a single variable.

With your class



14. How did we use data to determine the change in a single variable (i.e. Δt , Δv)?

With a partner



15. What do our equation(s) predict the velocity of the car will be after the truck hits it in a rear end collision? What is Δv for the car?

Add to your personal glossary

In your notebook



16. Take a moment to add to or record a new definition for the following terms in your personal glossary:

- Delta (Δ)
- Elastic collisions
- Inelastic collisions
- Momentum

Use mathematical thinking

In your notebook



17. Use our equations for momentum conservation to solve for unknown outcomes or conditions in three different collisions scenarios.

Self-assess and reflect

In your notebook



18. Use the key to self-assess your work.

19. Complete questions A-C to help identify part of the process you feel confident about and parts you'd like additional practice or help with.

20. Turn in both handouts when you are done.

Navigate

Turn and talk



21. How can we use what we figured out about change in momentum ($m\Delta V$) to make predictions about the connections to other variables we are interested in?

- F the force applied to an object
- Δt the time over which the force is applied